# TrustLayer

## Safe-by-Design AI Infrastructure for Healthcare

### Healthcare AI Product Case Study

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Healthcare AI systems are generating clinical content that directly affects patient care. The models behind them hallucinate, leak PHI, and occasionally produce unsafe guidance. TrustLayer is the safety infrastructure layer that catches these failures before they reach patients, clinicians, or regulated systems.

*All targets and thresholds in this document are proposed operating benchmarks for initial rollout, not validated production outcomes.*

# Contents

1. Executive Summary
2. Problem and Stakes
3. Product Thesis
4. Safety Control Architecture
5. Solution Structure
6. Metrics and Rollout
7. Risks
8. PM Takeaway

# 1. Executive Summary

Healthcare organizations are deploying AI into clinical workflows: patient messaging, clinical documentation, diagnostic support, prior authorization. The models are getting better, but they still hallucinate clinical facts, surface protected health information in the wrong context, and occasionally generate guidance that could cause harm. Most deployments have no independent safety layer between the model and the patient.

TrustLayer is a safety infrastructure layer that sits between healthcare AI models and clinical endpoints. Every AI output passes through a pipeline of checks: grounding verification, hallucination detection, PHI scanning, clinical safety classification, and policy evaluation. Based on these checks, the output is approved, escalated to a clinician, or blocked. Everything is logged in an immutable audit trail.

| Dimension | Detail |
|---|---|
| Core users | Clinical informatics, compliance, clinician reviewers |
| Buyer | CMIO, CISO, CIO / VP Digital Health |
| North star | Safe + Compliant Response Rate |
| Phase 1 scope | Grounding, hallucination, PHI, safety, escalation, audit |
| Phase 1 timeline | Months 1 to 5, single AI workflow, single department |
| Key risk | Hallucinated clinical content reaching a patient |

# 2. Problem and Stakes

Healthcare AI outputs are entering clinical workflows without independent safety validation. The specific failure modes are more dangerous than in other AI applications because the downstream consumer is a patient or a clinician making care decisions.

| Failure Mode | Example | Consequence |
|---|---|---|
| Hallucinated clinical content | AI fabricates a medication the patient does not take | Prescribing error based on false medication list |
| PHI leakage | Patient name appears in a response routed to wrong patient | HIPAA violation, potential $50K+ penalty |
| Unsafe clinical guidance | AI recommends a contraindicated treatment | Direct patient harm |
| No audit trail | Regulator asks what AI generated; no record exists | Regulatory compliance failure |

Current approaches are insufficient. Model-level safeguards reduce but do not eliminate unsafe outputs. Manual clinician review of all AI outputs does not scale. No validation at all is the most common and most dangerous current state.

# 3. Product Thesis

Healthcare AI safety is not a model problem. It is an infrastructure problem. You can improve models through fine-tuning, RLHF, and retrieval augmentation. That helps. But it does not eliminate the possibility of unsafe outputs. Healthcare needs a separate safety layer that catches what the model misses, operates independently, and is auditable.

This is the same principle behind pharmaceutical quality control: you do not rely on the manufacturing process alone. You have independent checks, operated by a different team, with their own audit trail. Healthcare AI needs the equivalent.

Why a dedicated safety layer over the alternatives:

| Dimension | Model Toolkit | Safety Layer | Review Platform |
|---|---|---|---|
| Catches model failures | Partially | Yes (independent) | Yes (human) |
| Independent audit | No | Yes | Partial |
| Scalability | With model | With service | Does not scale |
| Clinician burden | None | Low (escalation) | High (all outputs) |
| PHI protection | Model-dependent | Dedicated scanner | Human catches |
| Regulatory evidence | Weak | Strong | Strong but manual |

# 4. Safety Control Architecture

## Safety Pipeline

*Sequential safety checks applied to every AI output before clinical delivery.*

## Governance Principles

**Do No Harm by Default:** If the pipeline cannot confirm safety, the output does not reach the patient.

**Evidence-Based Validation:** Clinical claims must be traceable to source evidence.

**Privacy as a Hard Constraint:** PHI protection is zero-tolerance, not a configurable threshold.

**Clinician Authority:** Humans have the final say. Overrides are documented, not prevented.

**Full Traceability:** Every output, check, and decision is recorded and retrievable.

**Graceful Degradation:** If TrustLayer is unavailable, outputs are queued, not delivered.

| Disposition | Criteria | Behavior |
|---|---|---|
| Approved | All checks pass; risk below threshold | Output delivered to clinical endpoint |
| Escalated | Ambiguous findings; moderate risk | Routed to clinician with context and risk score |
| Blocked | Safety violation; high risk | Prevented; admin alerted; incident logged |

# 5. Solution Structure

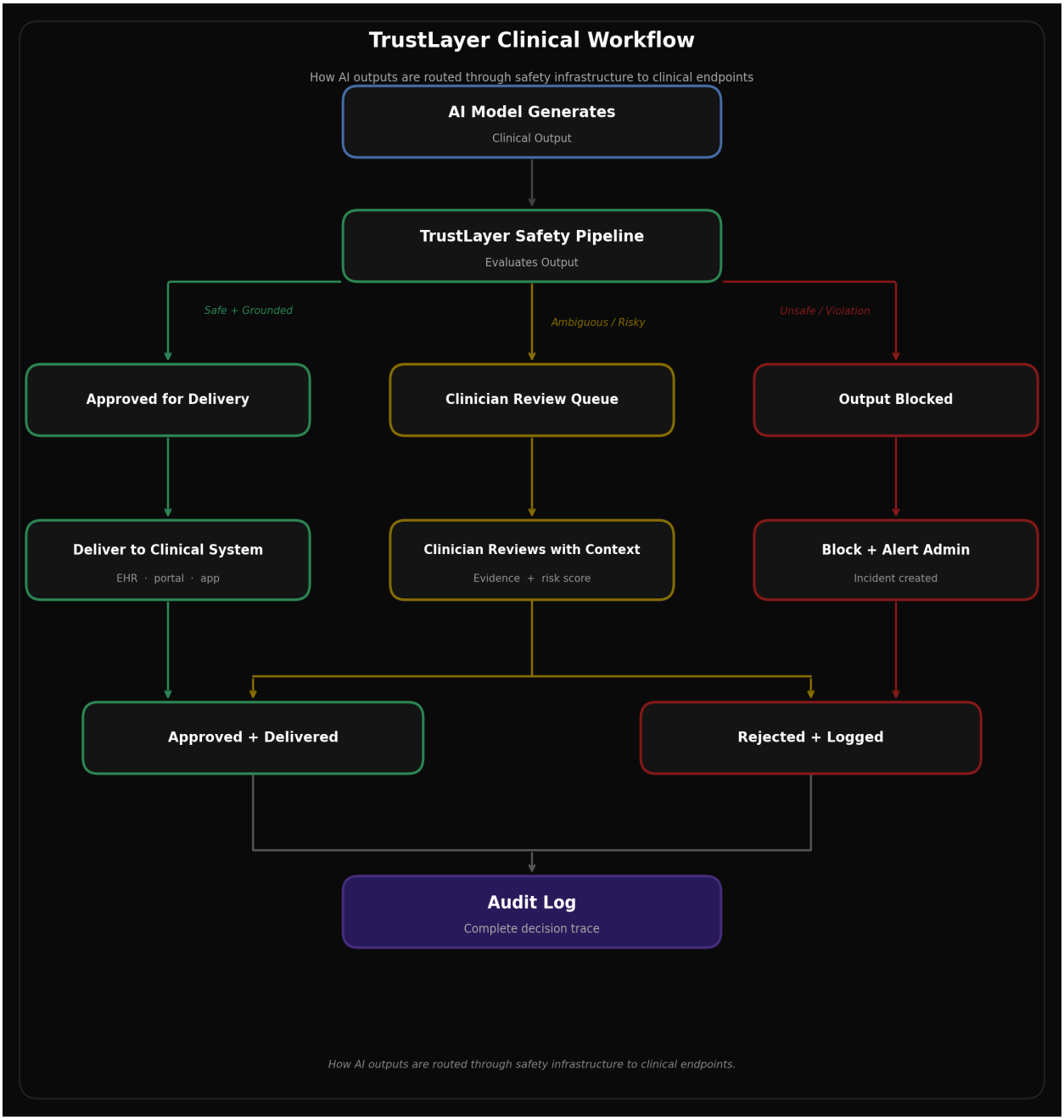
TrustLayer sits between AI models and clinical systems, intercepting every output for safety evaluation.

## Core Components

| Component | Purpose |
|---|---|
| Grounding Checker | Verifies clinical claims against patient records and guidelines |
| Hallucination Detector | Identifies fabricated medications, lab values, clinical facts |
| PHI Scanner | Detects all 18 HIPAA identifiers; healthcare-tuned entity recognition |
| Clinical Safety Classifier | Evaluates harm potential: dosage, contraindications, scope |
| Policy Engine | Configurable governance rules per workflow and department |
| Escalation Router | Routes ambiguous outputs to appropriate clinician reviewer |
| Audit Logger | Immutable trace of every output and every safety decision |
| Kill Switch | Infrastructure-level emergency halt, independent of application |

# Clinical Workflow

How AI outputs are routed through safety infrastructure to clinical endpoints.




*How AI outputs are routed through safety infrastructure to clinical endpoints.*

Every path through the workflow terminates at the audit log. Whether an output is approved, escalated and reviewed, or blocked, the complete decision context is recorded: source model, raw output, each safety check result, risk score, policy version, and final disposition.

# 6. Metrics and Rollout

Metrics hierarchy from north star through supporting, operational, and guardrail indicators.

## Proposed Targets (Initial Rollout)

*All targets are proposed benchmarks, to be calibrated from pilot data during shadow mode.*

| Metric | Proposed Target | Category |
|---|---|---|
| Safe + Compliant Response Rate | > 95% (launch) | North Star |
| Hallucination interception | > 99% | Supporting |
| PHI leakage incidents | 0 (zero tolerance) | Guardrail |
| Clinically unsafe miss rate | < 0.05% | Guardrail |
| Escalation precision | > 80% | Supporting |
| False-positive block rate | < 10% | Supporting |
| Safety pipeline latency | < 500ms (p99) | Guardrail |
| Audit coverage | 100% | Guardrail |
| Clinician override rate | Monitor (drift signal) | Operational |

## Phased Rollout

| Phase | Timeline | Deliverables |
|---|---|---|
| Phase 1: MVP | Mo 1–5 | Grounding, hallucination, PHI, safety classification, escalation, audit. Single workflow (patient messaging). |
| Phase 2: Expansion | Mo 6–10 | Configurable policy engine, dashboards, evidence validation, workflow-specific safety profiles. Multiple workflows. |
| Phase 3: Platform | Mo 11–16 | EHR integration (Epic, Oracle Health), adaptive thresholds, compliance reporting, multi-model governance. |

Each phase follows a shadow-mode, advisory-mode, enforcement-mode progression. Shadow mode runs the full pipeline without blocking, so the team can calibrate detection accuracy against real clinical outputs before enforcement begins.

# 7. Risks

Healthcare AI failure modes by likelihood and clinical impact.

| Risk | Severity | Mitigation |
|---|---|---|
| Hallucinated diagnosis reaches patient | **Critical** | Multi-method detection; conservative thresholds; mandatory shadow mode |
| PHI leakage in AI output | **Critical** | Zero-tolerance blocking; healthcare-tuned entity recognition |
| Unsafe dosage advice delivered | **Critical** | Drug DB cross-reference; dosage range checks; safety classifier |
| False-positive blocking at scale | **High** | Precision monitoring; clinician feedback loop; per-workflow calibration |
| Clinician review bottleneck | **High** | Escalation precision > 80%; queue management; SLA routing |
| Grounding failure on complex narratives | **Medium** | Iterative improvement with clinician feedback; known-limitation docs |
| Policy drift vs. clinical guidelines | **Medium** | Quarterly review; clinical advisory board; policy expiration dates |
| Audit gap in safety trace | **High** | Synchronous logging; cryptographic signing; daily integrity checks |

# 8. PM Takeaway

Healthcare AI safety is an infrastructure problem, not a model problem. You can improve models indefinitely and they will still produce unsafe outputs. Healthcare needs a separate layer that catches what the model misses, operates independently, and creates the audit evidence that regulators expect.

The design challenge is calibration. Healthcare has near-zero tolerance for certain failure modes (PHI leakage, unsafe dosage advice) but cannot practically operate if the safety layer blocks a large fraction of outputs. Finding the right sensitivity for each check, per workflow, per clinical context, is the core operating problem. It is also why a configurable policy engine is Phase 2 and not Phase 1: you need real data before you can build policies that generalize.

## Key Product Decisions

**Independent safety layer over model-level safeguards.** The same principle as pharmaceutical QC: independent checks operated by a different team with their own audit trail.

**Sequential pipeline over single-score classification.** Different failure modes require different detection methods. A hallucination detector and a PHI scanner solve different problems.

**Do-no-harm default over allow-by-default.** In healthcare, the cost of delivering an unsafe output far exceeds the cost of blocking a safe one.

**Clinician authority over system authority.** The safety pipeline provides signals. Clinical judgment belongs to the human. Overrides are documented, not prevented.

**Shadow mode before enforcement.** Detection accuracy must be validated against real clinical outputs before safety checks start blocking production content.

## Strategic Value

TrustLayer unlocks clinical AI adoption. Healthcare organizations that want to deploy AI but cannot accept the liability of unvalidated outputs need this infrastructure. It does not compete with AI model vendors. It makes their products deployable in regulated environments. The organization that defines how healthcare AI outputs are validated and governed establishes the safety standard that health systems adopt.

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